Week 10 Assignment

Multiple regression, ANCOVA, and interactions

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf`, along with your Data Safari submission, via Canvas before Week 11's lecture (Monday). See the syllabus for the late policy.

Note: This week also includes a Data Safari exercise — a separate, short assignment. Both are due at the same time.

```
library(tidyverse)
library(marginaleffects)
```

Part 1: Multiple regression

Using the built-in `trees` dataset:

1. Fit `lm(Volume ~ Girth + Height)` and `lm(Volume ~ Girth)`.
2. Compare the two models with `anova()`.
3. Use `avg_slopes()` on the two-predictor model. Interpret the `Girth` slope in one sentence, including its 95% confidence interval and units.

Part 2: ANCOVA and interactions

Using the built-in `iris` dataset, model `Petal.Width` as a function of `Petal.Length` and `Species`.

4. Fit the no-interaction (ANCOVA) model and the interaction model, then compare them with `anova()`.
5. Use `avg_slopes()` with `by = "Species"` to report the slope of `Petal.Length` within each species.
6. Make a plot of the interaction model's predictions with `plot_predictions()`.
7. In 2–3 sentences: does the slope depend on species? Support your answer with numbers from your outputs.

Part 3: Two-way design

Using `ToothGrowth` (convert `dose` to a factor first):

8. Fit `len ~ supp * dose`, then use `avg_comparisons()` with `by = "dose"` to estimate the supplement effect at each dose.

9. At which dose(s) does the supplement matter, and how large is the effect? Why can't the main effect of `supp` alone summarize this?